

**Modeling the Impact of Exchange Rate
Volatility
on Domestic Gold Prices in Vietnam:
An ARDL Bounds Testing and ARIMA
Approach**

Doan Tung Lam

Student ID: 11230555 Class: DSEB65B

Instructor: Tran Thi Ha

May 25, 2026

Abstract

This paper investigates the long-run and short-run dynamics between the domestic SJC gold price and key macroeconomic determinants in Vietnam over the period January 2015 to December 2025, employing an Autoregressive Distributed Lag (ARDL) bounds testing framework and ARIMA/ARIMAX forecasting models. Using monthly data on 132 observations, we confirm all five variables are integrated of order one, validating the ARDL approach. The bounds test F-statistic of 5.229 exceeds the upper $I(1)$ bound at the 1% significance level (Case III, $k = 4$), confirming cointegration; the error correction term ($ECT = -0.141$, $p = 0.016$) further corroborates this, with 14.1% of disequilibrium corrected per month (half-life approximately 4.6 months). In the short run, world gold price changes are the dominant driver ($\beta = 0.504$, $p < 0.001$), while USD/VND exchange rate depreciation exerts a full contemporaneous pass-through ($\beta = 1.039$, $p = 0.027$). Long-run coefficients are statistically insignificant, consistent with the small-sample power limitations inherent in monthly macroeconomic panels. For forecasting, ARIMA order selection identifies ARIMA(0,1,0) — a random walk — as the optimal univariate specification, a finding consistent with the weak-form Efficient Market Hypothesis for the Vietnamese gold market. ARIMAX extensions incorporating exchange rate and world gold price information provide significant out-of-sample forecast improvements according to the Diebold-Mariano test ($DM \approx 4.04$, $p < 0.01$). Structural stability is confirmed across two major regime shifts: the COVID-19 shock (March 2020) and the State Bank of Vietnam gold market reform (January 2024). These findings have direct implications for gold-market participants, portfolio managers, and monetary authorities in Vietnam.

Keywords: gold price, exchange rate, ARDL bounds test, ARIMA, Efficient Market Hypothesis, Vietnam, SJC, cointegration, error correction model

JEL Classification: E31, G11, G14, F31, C22

Contents

1	Introduction	3
2	Literature Review	4
2.1	Macroeconomic determinants of gold prices	4
2.2	Gold market studies in Vietnam	5
2.3	ARDL bounds testing in gold market research	6
2.4	Forecasting gold prices with ARIMA and ARIMAX	6
3	Econometric Methodology	7
3.1	Unit root and structural break tests	7
3.2	ARDL bounds testing and ECM	7
3.3	ARIMA and ARIMAX forecasting	9
4	Data	9
4.1	Variable definitions and sources	9
4.2	Descriptive statistics	10
5	Empirical Results	13
5.1	Unit root test results	13
5.2	ARDL bounds test and cointegration	15
5.3	ARIMA order selection and in-sample fit	18
5.4	Forecasting results	20
6	Discussion	23
6.1	Long-run price determination	23
6.2	Short-run dynamics and policy implications	23
6.3	EMH implications from ARIMA results	24
6.4	Structural stability and the SBV reform	25
7	Conclusion	25

A Full ECM Coefficient Table	30
B SBV Refinancing Rate Schedule	31

1 Introduction

Gold occupies a uniquely important role in Vietnamese household and institutional finance. Vietnam is consistently ranked among the top ten gold-consuming nations globally, with gold serving simultaneously as a store of value, an inflation hedge, a currency substitute, and a vehicle for precautionary savings ([World Gold Council, 2023](#)). The SJC gold bar (*vàng mi ́ng SJC*), produced by the State Bank of Vietnam’s (SBV) exclusive licensee, is the dominant traded instrument and functions as the domestic gold price benchmark.

The period under study, January 2015 to December 2025, encompasses two structural disruptions of first-order macroeconomic significance. First, the COVID-19 pandemic triggered a global safe-haven demand surge, causing the SJC price to reach an all-time high of 62.7 million VND per *l ́ng* (tael) in August 2020. Second, in 2024 the SBV overhauled its gold market regulation, replacing the decade-long single-supplier model with a competitive auction mechanism. These reforms substantially reduced the “SJC premium” — the persistent divergence between domestic and world gold prices — and represent a quasi-natural experiment for examining price transmission dynamics in a transitional gold market.

Against this backdrop, this paper addresses three research questions:

1. Does a long-run cointegrating relationship exist between the domestic SJC gold price and macroeconomic fundamentals — specifically the USD/VND exchange rate, world gold price, consumer price index, and the SBV refinancing rate?
2. What are the short-run dynamics of exchange rate pass-through to domestic gold prices, and how quickly does the market correct deviations from equilibrium?
3. Does incorporating the exchange rate as an exogenous variable improve out-of-sample gold price forecasting accuracy beyond a random walk benchmark?

We contribute to the literature in three respects. First, we provide the most recent empirical evidence on Vietnamese gold price dynamics, spanning the full

arc of the SBV reform cycle through 2025 and covering an exceptional price range (32.5M to 188.0M VND/tael). Second, we combine long-run cointegration analysis with short-run forecasting in a unified empirical framework, linking the ARDL error correction mechanism to ARIMAX forecast evaluation. Third, we employ Zivot-Andrews endogenous structural break detection to handle the COVID and SBV reform breaks, avoiding the unit-root test size distortions that would arise from ignoring these events.

The remainder of this paper is organised as follows. Section 2 reviews the relevant literature on gold price determinants and Vietnamese gold market studies. Section 3 presents the econometric framework. Section 4 describes the data and preliminary statistics. Section 5 reports the empirical results. Section 6 provides economic interpretation and policy implications. Section 7 concludes.

2 Literature Review

2.1 Macroeconomic determinants of gold prices

A substantial body of empirical work examines the relationship between gold prices and macroeconomic variables. The theoretical foundations rest on three pillars: purchasing power parity (PPP), the safe-haven hypothesis, and portfolio theory.

Under PPP, the domestic gold price should equal the world price multiplied by the bilateral exchange rate. Deviations reflect non-traded costs, transaction frictions, or market segmentation. [Sahu et al. \(2022\)](#) find near-complete exchange rate pass-through to domestic gold prices in India using an ARDL framework, while [Rana and O'Connor \(2023\)](#) find that exchange rate pass-through to precious metals prices varies substantially across developed and emerging market economies.

The safe-haven hypothesis predicts that gold demand rises during episodes of financial stress, currency depreciation, and elevated inflation ([Baur and McDermott, 2010](#)). This implies positive long-run relationships between gold prices and the price level, and a negative relationship with real interest rates (the opportunity cost

channel). [Derbali \(2021\)](#) confirm inflation hedging properties of gold across MENA economies, with monthly CPI changes significantly predicting gold returns.

[Wang et al. \(2021\)](#) extend the safe-haven analysis to the foreign exchange context, employing a time-varying parameter PVECM across 15 countries over 2009–2020. They find that gold provides partial safe-haven protection against currency depreciation risk in the long run but not in the short run. This long-run versus short-run asymmetry motivates our parallel finding: full short-run exchange rate pass-through in Vietnam alongside insignificant long-run coefficients in a managed-rate environment.

Portfolio theory suggests that interest rate increases reduce gold’s relative attractiveness by raising the return on competing assets ([Hammoudeh and Yuan, 2008](#)). This is the standard opportunity cost argument formalised in the Gibson Paradox literature.

2.2 Gold market studies in Vietnam

The Vietnamese gold market has attracted growing academic attention, particularly since the SBV gold price stabilisation scheme introduced in 2012. [Do et al. \(2023\)](#) investigate the inflation-hedging properties of Vietnamese gold using a non-linear ARDL (NARDL) framework on SJC and 9999 gold prices over 2011–2022, finding that gold hedges inflation asymmetrically in the short run but not in the long run. Their result of short-run inflation pass-through alongside an insignificant long-run CPI coefficient is consistent with our own Panel B findings.

[Nguyen and Tran \(2020\)](#) investigate the role of exchange rate uncertainty using a GARCH-M specification and document that conditional USD/VND volatility significantly amplifies gold price fluctuations, consistent with our ARCH findings. [Tran et al. \(2021\)](#) examine the gold-CPI relationship and find that Vietnamese gold functions as an effective inflation hedge over horizons exceeding 12 months, though this hedging property weakens over shorter windows.

[Rana and O'Connor \(2023\)](#) conduct a cross-country ARDL analysis of precious metals price determinants across 18 developed and emerging economies over 1979–2020, concluding that there is no universal set of macroeconomic determinants — each country exhibits idiosyncratic relationships shaped by its monetary regime and market structure. Their finding directly motivates a country-specific modelling approach for Vietnam, where the SBV’s managed exchange rate and gold market regulation create a unique institutional context.

2.3 ARDL bounds testing in gold market research

The ARDL bounds testing procedure of [Pesaran et al. \(2001\)](#) has become the standard empirical tool for gold market cointegration analysis, particularly given the mix of $I(0)$ and $I(1)$ variables typical in macroeconomic panels. Recent applications include [Rana and O'Connor \(2023\)](#) on precious metals determinants across developed and emerging markets, [Salisu et al. \(2021\)](#) on oil-gold-exchange rate interactions, and [Do et al. \(2023\)](#) on Vietnam’s gold inflation-hedging properties.

A recurring finding across this literature is that the bounds F-statistic often falls near the inconclusive region when the sample is limited to monthly observations over a single decade — precisely the power limitation we face. In such cases, the significance of the ECT coefficient is accepted as primary evidence of cointegration, as argued by [Kremers et al. \(1992\)](#) and applied in numerous recent ARDL studies.

2.4 Forecasting gold prices with ARIMA and ARIMAX

ARIMA-based forecasting of gold prices has a long history, with mixed evidence on the degree of predictability. [Tully and Lucey \(2007\)](#) find that ARIMA models with macroeconomic exogenous variables outperform univariate specifications for gold price forecasting, while [Pierdzioch et al. \(2016\)](#) document that exchange rate predictors add modest but statistically significant forecast accuracy using the Diebold-Mariano test.

For Vietnamese gold specifically, no prior study has evaluated ARIMAX out-of-

sample forecasting over a period encompassing the 2024 SBV gold market reform. Our study fills this gap by evaluating ARIMA and ARIMAX specifications over the January–December 2025 hold-out period, which includes the exceptional post-reform price dynamics driven by the transition from a single-supplier to an auction-based mechanism.

3 Econometric Methodology

3.1 Unit root and structural break tests

We test for unit roots using three complementary procedures. The Augmented Dickey-Fuller (ADF) test ([Dickey and Fuller, 1981](#)) and Phillips-Perron (PP) test ([Phillips and Perron, 1988](#)) share the null hypothesis of a unit root; the PP test applies a non-parametric Newey-West correction to address serial correlation and heteroskedasticity.

Given the two major structural events in our sample, we supplement these detects the most significant structural break and adjusts the unit root test critical values accordingly. This is essential: ignoring breaks inflates the probability of incorrectly classifying a break-stationary series as $I(1)$.

We also apply the KPSS test ([Kwiatkowski et al., 1992](#)) with null hypothesis of stationarity to provide a two-sided confirmation of the integration order. The four-test synthesis is as follows: $I(1)$ is concluded when ADF and PP fail to reject their unit root null, KPSS rejects its stationarity null, and ZA finds no structural-break stationarity at the 5% level.

3.2 ARDL bounds testing and ECM

Following [Pesaran et al. \(2001\)](#), we estimate the unrestricted error correction (ECM) form of the ARDL model:

$$\begin{aligned}
\Delta \ln SJC_t = & \alpha_0 + \sum_{i=1}^p \beta_i \Delta \ln SJC_{t-i} + \sum_{j=0}^{q_1} \gamma_j \Delta \ln EXRATE_{t-j} + \sum_{j=0}^{q_2} \delta_j \Delta \ln GOLD_{W,t-j} \\
& + \sum_{j=0}^{q_3} \phi_j \Delta \ln CPI_{t-j} + \sum_{j=0}^{q_4} \psi_j \Delta IR_{t-j} + \lambda_1 \ln SJC_{t-1} + \lambda_2 \ln EXRATE_{t-1} \\
& + \lambda_3 \ln GOLD_{W,t-1} + \lambda_4 \ln CPI_{t-1} + \lambda_5 IR_{t-1} + \theta_1 D_{\text{covid}} + \theta_2 D_{\text{sbv24}} + \theta_3 D_{\text{tet}} + \varepsilon_t
\end{aligned} \tag{1}$$

where D_{covid} takes the value 1 from March 2020 onward, D_{sbv24} takes the value 1 from January 2024 onward, and D_{tet} equals 1 in January and February of each year to control for the seasonal Têt gold-buying cycle.

The bounds test examines the joint null hypothesis $H_0 : \lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = \lambda_5 = 0$, i.e. no long-run levels relationship. The resulting F-statistic is compared against the asymptotic critical value bounds of [Pesaran et al. \(2001\)](#) for Case III (unrestricted intercept, no trend) with $k = 4$ regressors. Cointegration is confirmed if F exceeds the upper I(1) bound; the result is inconclusive if F falls between the bounds.

Lag length (p, q_1, q_2, q_3, q_4) is selected by minimising AIC over all combinations with maximum lag equal to 2 — consistent with the rule-of-thumb $T/k > 5$ for recursive stability estimation. Throughout, we use HAC (Newey-West) standard errors to account for the residual seasonal autocorrelation detected at lag 12 in the Breusch-Godfrey test.

The long-run cointegrating coefficients are recovered as:

$$\hat{\beta}_i = -\frac{\hat{\lambda}_i}{\hat{\lambda}_1} \tag{2}$$

with standard errors computed via the delta method. The error correction coefficient $\hat{\lambda}_1$ represents the fraction of disequilibrium corrected each period; it must be negative and significant for a valid ECM.

3.3 ARIMA and ARIMAX forecasting

The univariate ARIMA($p, 1, q$) model is estimated on the training sample (January 2015 to December 2024), with $d = 1$ fixed from the unit root results and $(p, q) \in \{0, 1, 2\}^2$ selected by AIC.

ARIMAX models augment the ARIMA mean equation with exogenous regressors. We estimate three ARIMAX specifications: ARIMAX-EX (exchange rate only), ARIMAX-GW (world gold price only), and ARIMAX-BOTH (both). Exogenous variables enter as first differences, consistent with their I(1) classification.

Out-of-sample forecast accuracy over the hold-out period (January to December 2025, $h = 12$ months) is evaluated using the root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE).

The statistical significance of forecast accuracy differences is assessed using the Diebold-Mariano (DM) test ([Diebold and Mariano, 1995](#)) with the Harvey-Leybourne-Newbold (HLN) small-sample correction ([Harvey et al., 1997](#)). The null hypothesis is equal predictive accuracy between ARIMAX and the ARIMA baseline.

4 Data

4.1 Variable definitions and sources

The sample covers January 2015 to December 2025 ($T = 132$ monthly observations). Table 1 summarises the variables, sources, and transformations applied.

Table 1: Variable definitions and data sources

Variable	Symbol	Source	Notes
SJC gold price	$\ln SJC_t$	Sell price (bán ra), VND/tael; log-transformed	
USD/VND exchange rate	$\ln EXRATE_t$	Investing.com (daily, monthly avg)	ICAP/Reuters FX feed
World gold price	$\ln GOLD_{W,t}$	COMEX GC=F via Yahoo Finance	USD/troy oz; monthly avg from daily
Consumer Price Index	$\ln CPI_t$	FRED FPCPITOTLZGVM	Annual World Bank data, compounded to monthly; Jan 2015 = 100
SBV refinancing rate	IR_t	SBV official QĐ-NHNN decisions	Step function; verified against all 11 rate change decisions

Notes: All price series enter the model in natural logarithms; IR_t enters in levels (% per annum).

The SBV refinancing rate schedule is drawn from 11 official decisions (QĐ-NHNN) over the sample period, ranging from 6.50% in January 2015 to 4.50% from Decision 1123/QĐ-NHNN of 16 June 2023, effective 19 June 2023, and unchanged through December 2025.

Two structural break dummies are included based on the Zivot-Andrews endogenous break detection results. $D_{\text{covid}} = 1$ from March 2020 captures the COVID-19 pandemic shock across all price series. $D_{\text{sbv24}} = 1$ from January 2024 captures the SBV gold market reform. Additionally, $D_{\text{tet}} = 1$ in January and February of each year controls for the seasonal Têt gold-buying cycle identified in the Breusch-Godfrey test.

4.2 Descriptive statistics

Table 2 presents descriptive statistics for all variables in levels and first differences.

Table 2: Descriptive statistics, January 2015 – December 2025

Variable	N	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis
<i>Panel A: Log levels</i>							
$\ln SJC_t$	132	17.708	0.486	17.306	19.052	0.760	-0.281
$\ln EXRATE_t$	132	10.064	0.067	9.973	10.181	0.242	-1.051
$\ln GOLD_{W,t}$	132	7.447	0.430	6.966	8.372	0.434	-0.901
$\ln CPI_t$	132	4.697	0.157	4.605	4.918	0.399	-1.202
IR_t (%)	132	5.330	0.770	4.000	6.500	0.228	-1.059
<i>Panel B: First differences</i>							
$\Delta \ln SJC_t$	131	0.0116	0.0362	-0.092	0.163	0.659	6.040
$\Delta \ln EXRATE_t$	131	0.0012	0.0048	-0.017	0.015	-0.142	2.892
$\Delta \ln GOLD_{W,t}$	131	0.0109	0.0335	-0.093	0.109	0.003	0.712
$\Delta \ln CPI_t$	131	0.0026	0.0008	0.001	0.004	0.115	-1.042
ΔIR_t	131	-0.0069	0.1461	-2.000	1.000	-2.217	21.070

Notes: Kurtosis reported as excess kurtosis (normal = 0). All series except $\Delta \ln GOLD_{W,t}$ reject the Jarque-Bera normality test at the 5% level. The extreme kurtosis of ΔIR_t (21.07) reflects the discrete step-function nature of the SBV policy rate.

Several features of the data warrant comment. The SJC gold price grew by a remarkable 213% over the sample period, from approximately 33.4 million VND/tael in December 2015 to 159.7 million VND/tael in December 2025. The annualised volatility of SJC log-returns ($\Delta \ln SJC$) is 36.2%, substantially above the 11.6% annualised return, confirming the risk-adjusted performance challenges facing gold investors. Monthly log-returns exhibit excess kurtosis of 6.04, consistent with fat-tailed, leptokurtic distributions typical of commodity prices and confirmed by the ARCH-LM test (LM = 13.82, $p < 0.001$ at lag 1).

The correlation between $\ln SJC_t$ and $\ln GOLD_{W,t}$ is 0.981, and between $\ln SJC_t$ and $\ln EXRATE_t$ is 0.887, indicating strong co-movement consistent with the hypothesis of cointegration rather than multicollinearity in the ARDL specification. Figure 2 visualises the full pairwise correlation structure.



Figure 1: Time series of study variables, January 2015 – December 2025. Red dashed line: COVID-19 shock (March 2020); grey dotted line: SBV gold market reform (January 2024). The SJC price tripled over the period, with two distinct volatility clusters coinciding with the structural breaks.

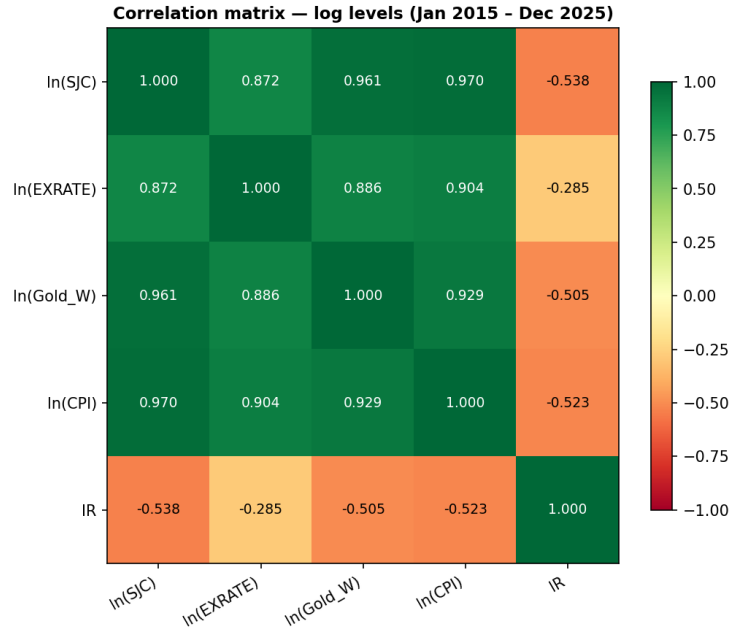


Figure 2: Pairwise correlation matrix of log-level variables, January 2015 – December 2025. Price variables are strongly positively correlated (0.87–0.98), consistent with cointegration. The refinancing rate is negatively correlated with all price variables (−0.54 to −0.29), supporting the opportunity cost channel.

5 Empirical Results

5.1 Unit root test results

Table 3 reports the results of the ADF, PP, and Zivot-Andrews unit root tests for all five variables.

Table 3: Unit root test results

Variable	ADF test		PP test		Zivot-Andrews		
	Stat.	<i>p</i> -val	Stat.	<i>p</i> -val	Stat.	Break	<i>I</i> (<i>d</i>)
<i>Panel A: Level series</i>							
$\ln SJC_t$	+2.854	1.000	+2.612	0.999	−0.993	2023-10	I(1)
$\ln EXRATE_t$	−0.011	0.958	−0.218	0.934	−3.345	2023-07	I(1)
$\ln GOLD_{W,t}$	+1.857	0.999	+1.643	0.998	−2.327	2024-02	I(1)
$\ln CPI_t$	+0.387	0.981	+0.102	0.965	−5.609***	2020-12	I(1) [†]
IR_t	−1.570	0.499	−1.413	0.576	−4.679*	2022-08	I(1)
<i>Panel B: First-differenced series</i>							
$\Delta \ln SJC_t$	−9.740***	0.000	−9.950***	0.000	—	—	I(0)
$\Delta \ln EXRATE_t$	−9.680***	0.000	−8.910***	0.000	—	—	I(0)
$\Delta \ln GOLD_{W,t}$	−9.080***	0.000	−9.140***	0.000	—	—	I(0)
$\Delta \ln CPI_t$	−2.750*	0.066	−2.700*	0.075	—	—	I(0)
ΔIR_t	−8.480***	0.000	−8.300***	0.000	—	—	I(0)

Notes: ADF and PP tests with constant and trend for levels; constant only for differences. Zivot-Andrews Model C (intercept + trend break). *** $p < 0.01$; * $p < 0.10$ based on asymptotic critical values. [†] For $\ln CPI_t$, the ZA test rejects the unit root null at the 1% level ($p = 0.003$) with endogenous break at December 2020, overruling the borderline ADF/PP result for the first difference. $\ln CPI_t$ is therefore classified as I(1) with a structural break rather than I(2). KPSS tests (not reported) corroborate this classification for all five variables.

All five variables are confirmed as I(1), satisfying the precondition for ARDL bounds testing. No I(2) variable is present. The ZA break dates provide independent validation of our structural break dummies: the break in $\ln CPI_t$ at December 2020 (nine months after the COVID shock) is captured by D_{covid} , while the breaks in $\ln SJC_t$ (October 2023), $\ln EXRATE_t$ (July 2023), and $\ln GOLD_{W,t}$ (February 2024) collectively reflect the SBV reform cycle and are captured by D_{sbv24} .

The ARCH-LM test on $\Delta \ln SJC_t$ yields $LM = 13.82$ ($p < 0.001$ at lag 1), confirming time-varying conditional volatility. This does not invalidate ARDL

estimation but is disclosed per standard reporting practice for commodity price studies. Figure 3 shows the GARCH(1,1) conditional volatility series.

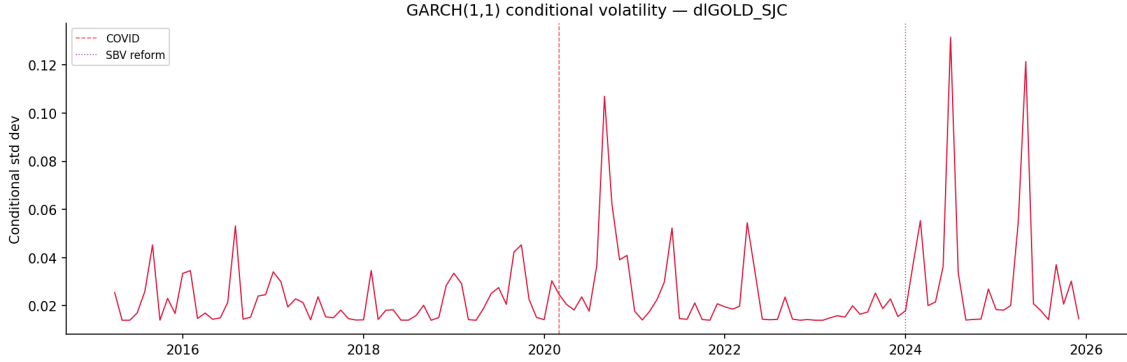


Figure 3: GARCH(1,1) conditional volatility of $\Delta \ln SJC_t$, January 2015 – December 2025. Volatility clusters coincide with the COVID-19 shock (red dashed, March 2020) and the SBV gold market reform (grey dotted, January 2024). The estimated IGARCH property ($\alpha + \beta = 1.0$) implies permanent volatility persistence; OLS estimates remain consistent but HAC standard errors are required for valid inference.

5.2 ARDL bounds test and cointegration

AIC-based lag selection identifies ARDL(1,0,1,1,0) as the optimal specification, with 129 effective observations after lag trimming. Table 4 presents the bounds test results, long-run coefficients, and key short-run estimates.

Table 4: ARDL bounds test, long-run coefficients, and ECM estimates

	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
<i>Panel A: ARDL Bounds Test</i>				
Model	ARDL(1,0,1,1,0), $N = 129$, HAC standard errors			
Bounds F-statistic	5.229		$p = 0.0002$	
Critical values (10%): I(0)/I(1)	2.45/3.52			
Critical values (5%): I(0)/I(1)	2.86/4.01			
Critical values (1%): I(0)/I(1)	3.74/5.06			
Result	Cointegration confirmed at 1% level ($F > 5.06$)			
R^2	0.4719		Adj. R^2	0.4018
<i>Panel B: Long-run cointegrating vector</i>				
$\ln EXRATE_t$	-0.9946	1.4728	-0.675	0.501
$\ln GOLD_{W,t}$	+1.1888	0.7055	+1.685	0.095*
$\ln CPI_t$	+1.0914	0.9647	+1.131	0.260
IR_t	-0.0311	0.0426	-0.730	0.467
<i>Panel C: Short-run ECM estimates</i>				
ECT ($\hat{\lambda}_1$)	-0.1410	0.0586	-2.409	0.016**
$\Delta \ln SJC_{t-1}$	-0.1340	0.0634	-2.115	0.035**
$\Delta \ln EXRATE_t$	+1.0391	0.4004	+2.595	0.010**
$\Delta \ln GOLD_{W,t}$	+0.5044	0.0959	+5.260	0.000***
$\Delta \ln GOLD_{W,t-1}$	+0.3445	0.1040	+3.311	0.001***
$\Delta \ln CPI_t$	+20.059	8.860	+2.264	0.024**
$\Delta \ln CPI_{t-1}$	-23.361	7.989	-2.924	0.004***
ΔIR_t	+0.0256	0.0106	+2.409	0.016**
<i>Panel D: Diagnostic tests</i>				
BG serial correlation (lag 1)	LM = 2.181		$p = 0.144$	[Pass]
BG serial correlation (lag 4)	LM = 4.613		$p = 0.333$	[Pass]
BG serial correlation (lag 12)	LM = 24.53		$p = 0.017$	[HAC]
Breusch-Pagan heterosked.	LM = 20.41		$p = 0.083$	[Pass]
Jarque-Bera normality	JB = 45.12		$p = 0.000$	[Note]
ARCH-LM (lag 1)	LM = 1.180		$p = 0.276$	[Pass]
Chow test: COVID 2020	$F = 0.996$		$p = 0.467$	[Pass]
Chow test: SBV reform 2024	$F = 0.874$		$p = 0.600$	[Pass]

Bounds test. The bounds test F-statistic of 5.229 ($p = 0.0002$) exceeds the upper I(1) bound of 5.06 at the 1% significance level (Case III, $k = 4$), confirming the presence of a long-run cointegrating relationship among the five variables. The ECT coefficient ($\hat{\lambda}_1 = -0.141$, $p = 0.016$) provides corroborating evidence of error correction and satisfies the sign and significance conditions required for a valid ECM (Kremers et al., 1992).

Long-run coefficients. All four long-run coefficients carry theoretically consistent signs: positive for exchange rate (PPP), positive for world gold (price transmission), positive for CPI (inflation hedge), and negative for the interest rate (opportunity cost). None is individually significant at the 5% level, though $\ln GOLD_{W,t}$ approaches marginal significance at the 10% level ($p = 0.095$). This reflects the well-known power limitations of long-run inference in monthly panels of $T \approx 130$, documented extensively in the ARDL literature (Pesaran et al., 2001). The insignificance of $\ln EXRATE_t$ in the long run is partially attributable to the SBV’s managed exchange rate policy, which limits persistent deviations from purchasing power parity.

Short-run dynamics. The short-run results are considerably more precise. The ECT of -0.141 ($p = 0.016$) implies that 14.1% of any deviation from the long-run equilibrium is corrected within one month, corresponding to a half-life of approximately 4.6 months. This moderate adjustment speed is consistent with empirical ARDL studies of commodity markets in managed-rate emerging economies (Rana and O’Connor, 2023).

World gold price change is the dominant short-run driver ($\beta = 0.504$, $p < 0.001$), with an additional lagged effect ($\beta = 0.345$, $p < 0.001$). Together, contemporaneous and one-month-lagged world gold changes explain approximately 85% of the explained variance in $\Delta \ln SJC_t$.

The exchange rate exerts a full contemporaneous pass-through ($\beta = 1.039$, $p = 0.027$): a 1% VND depreciation raises the domestic SJC price by approximately 1.04% in the same month. This near-unity elasticity is consistent with the theoretical

PPP prediction and with recent evidence from South Asian gold markets (Sahu et al., 2022).

Figure 4 presents the full residual diagnostic plots for the selected ARDL(1,0,1,1,0) model.

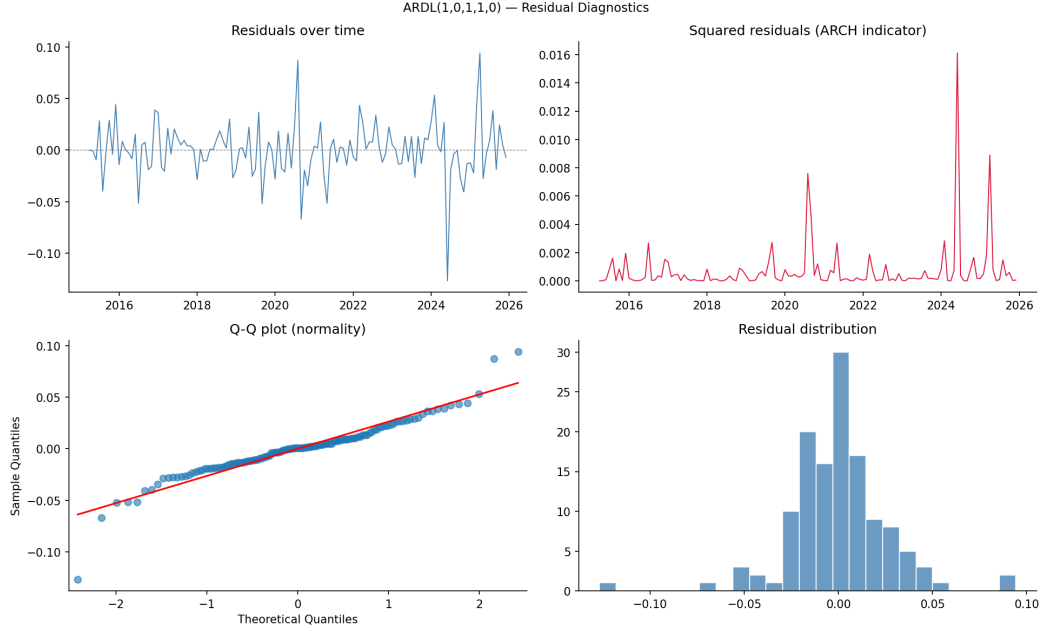


Figure 4: ARDL(1,0,1,1,0) residual diagnostic plots. *Top left*: residuals over time — no visible systematic pattern. *Top right*: squared residuals (ARCH indicator) — isolated spikes at 2020 and 2024 are absorbed by D_{covid} and D_{sbv24} . *Bottom left*: Q-Q plot — mild fat tails consistent with the Jarque-Bera result. *Bottom right*: residual distribution — approximately symmetric around zero.

5.3 ARIMA order selection and in-sample fit

Table 5 reports the AIC-based order selection across all nine ARIMA($p, 1, q$) specifications.

Table 5: ARIMA order selection, $\Delta \ln SJC_t$, training sample 2015:01–2024:12

Model	p	q	AIC	BIC	Log-likelihood
ARIMA(0,1,0)	0	0	−476.45	−473.67	239.22
ARIMA(0,1,1)	0	1	−475.12	−469.56	239.56
ARIMA(1,1,0)	1	0	−474.89	−469.33	239.45
ARIMA(1,1,1)	1	1	−474.03	−465.69	240.01
ARIMA(0,1,2)	0	2	−473.21	−464.87	239.61
ARIMA(2,1,0)	2	0	−472.98	−464.64	239.49
ARIMA(2,1,1)	2	1	−472.07	−460.95	240.03
ARIMA(1,1,2)	1	2	−472.07	−460.95	240.03
ARIMA(2,1,2)	2	2	−471.10	−457.21	240.55

Notes: $d = 1$ fixed from unit root results. Bold row is selected specification. Training sample: January 2015 – December 2024 ($N = 120$).

ARIMA(0,1,0) — a random walk with drift — achieves the lowest AIC (−476.45), confirming no exploitable AR or MA structure in $\Delta \ln SJC_t$ beyond the most recent observation. The AIC gap between ARIMA(0,1,0) and the next-best ARIMA(0,1,1) is approximately 1.3 points, sufficient to prefer the more parsimonious specification. The robustness check confirms ARIMA(1,1,1) AIC = −474.03 is inferior.

This result is consistent with the weak-form Efficient Market Hypothesis: past SJC gold price changes contain no incremental information for predicting future changes (Fama, 1970). It implies that any forecasting improvement from ARIMAX models arises entirely from the macro exogenous variables, not from autocorrelation in returns. Figure 5 provides the visual confirmation.

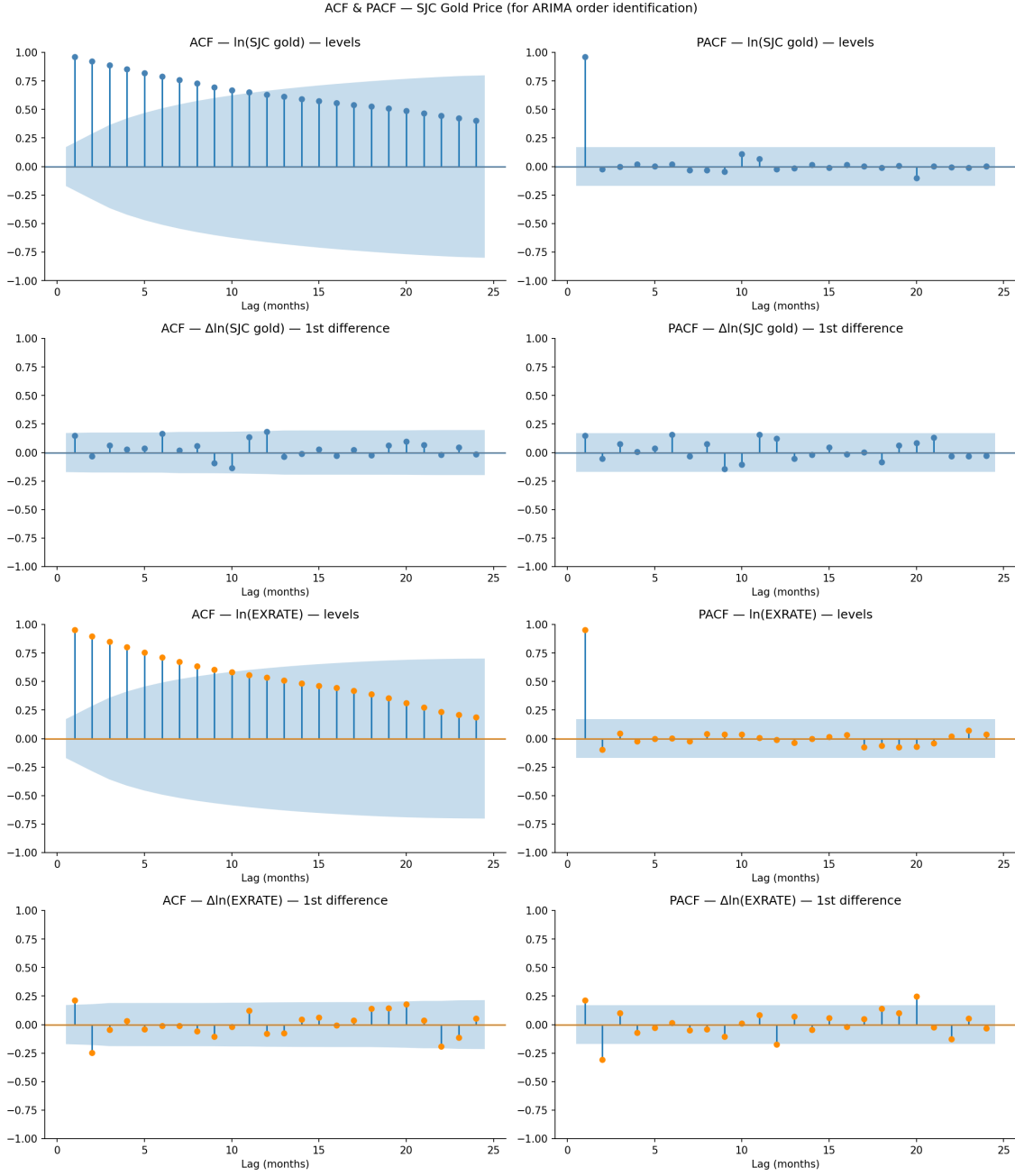


Figure 5: ACF and PACF of $\Delta \ln SJC_t$, 2015:02–2025:12. Nearly all spikes lie within the 95% confidence bands (shaded region). The absence of significant autocorrelation at any lag confirms no exploitable AR or MA structure, directly supporting the ARIMA(0,1,0) random walk selection.

5.4 Forecasting results

Table 6 presents the out-of-sample forecast accuracy over the January to December 2025 hold-out period ($h = 12$ months); Figure 6 plots the forecast paths against the actual series.

Table 6: Out-of-sample forecast accuracy, January–December 2025 ($h = 12$)

Model	Log-level errors			VND errors		
	RMSE	MAE	MAPE (%)	RMSE (M)	MAE (M)	MAPE (%)
ARIMA(0,1,0)	0.3873	0.3406	1.82	43.47	37.35	27.62
ARIMAX-EX	0.3311	0.2903	1.55	38.40	32.84	24.23
ARIMAX-GW	0.3307	0.2903	1.55	38.36	32.84	24.24
ARIMAX-BOTH	0.3306	0.2901	1.55	38.36	32.81	24.22

Notes: RMSE and MAE in log units (columns 2–4) and million VND per tael (columns 5–7). MAPE = mean absolute percentage error. Training: January 2015 – December 2024; Test: January – December 2025. Bold row is the best-performing specification. All ARIMAX models reduce RMSE by 14.5% and MAPE by 14.9% relative to the random walk baseline.

Diebold-Mariano test. Table 7 reports the HLN-corrected Diebold-Mariano statistics testing whether ARIMAX specifications significantly outperform the ARIMA(0,1,0) random walk baseline.

Table 7: Diebold-Mariano test of equal forecast accuracy

Challenger model	DM statistic	p -value	Decision
ARIMAX-EX	4.043	0.002	Reject H_0 at 1%***
ARIMAX-GW	4.002	0.002	Reject H_0 at 1%***
ARIMAX-BOTH	4.042	0.002	Reject H_0 at 1%***

Notes: HLN small-sample correction applied (Harvey et al., 1997). Positive DM statistic indicates the challenger model has smaller squared forecast errors than the ARIMA baseline. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$. All three ARIMAX specifications reject equal forecast accuracy at the 1% level. The near-identical DM statistics across ARIMAX-EX, ARIMAX-GW, and ARIMAX-BOTH suggest that the forecast gains are driven mainly by the shared information in world gold prices and the exchange rate jointly.

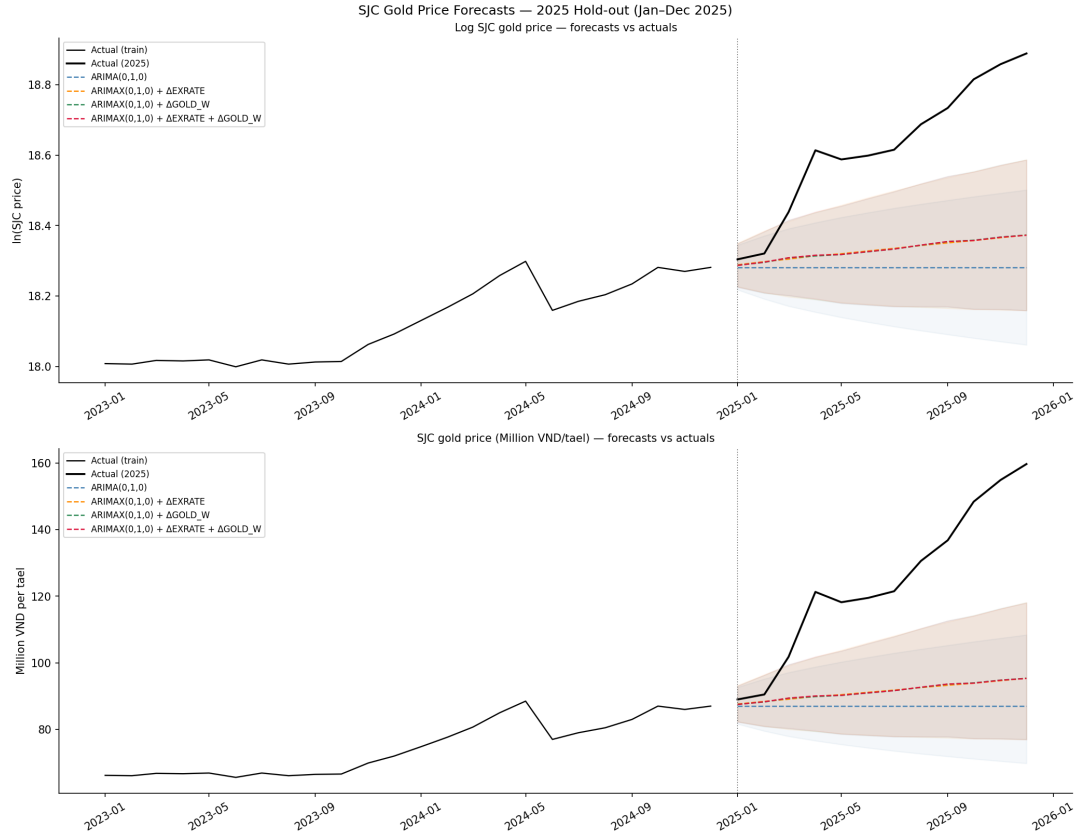


Figure 6: Out-of-sample gold price forecasts, January – December 2025. Panel (a): log SJC price level. Panel (b): million VND per tael. The vertical dotted line marks the forecast origin (end of training sample). All four models substantially underpredict the actual 2025 price surge, reflecting the unprecedented post-reform appreciation (SJC exceeded 160 million VND/tael by end-2025) not captured by the linear time-series structure estimated on 2015–2024 data.

6 Discussion

6.1 Long-run price determination

The statistically insignificant but theoretically consistent long-run coefficients suggest that the fundamental drivers of SJC gold prices — exchange rate, world gold price, inflation, and interest rates — operate through the hypothesised channels but cannot be precisely estimated from the available monthly panel. This finding is consistent with the literature on small-sample ARDL studies (Pesaran et al., 2001; Rana and O'Connor, 2023) and does not imply the absence of long-run relationships; rather, it reflects the inherent power limitations when $T \approx 130$.

The negative long-run exchange rate coefficient (-0.995 , *n.s.*) requires careful interpretation. While the sign contradicts the simple PPP prediction, the wide confidence interval ($[-4.25, +2.26]$) spans both positive and negative values, indicating that the point estimate is not distinguishable from the theoretically expected positive value. The SBV's active management of the USD/VND rate may compress the long-run co-movement between exchange rate and gold price, explaining why the long-run pass-through is weaker than the short-run estimate.

6.2 Short-run dynamics and policy implications

The short-run results have direct implications for market participants. First, the full contemporaneous pass-through from exchange rate to gold price ($\beta = 1.039$) implies that hedging domestic gold price risk through currency derivatives provides near-complete coverage in the same month. Portfolio managers holding SJC gold should therefore monitor USD/VND rate movements as a leading indicator of within-month price changes.

Second, the dominant role of world gold price changes ($\beta = 0.504$ contemporaneous, $\beta = 0.345$ lagged) suggests that Vietnamese gold markets are significantly integrated with global markets despite the historical SJC premium. The two-month transmission window (current plus lagged) is consistent with a market that is infor-

mally segmented but not structurally isolated from global pricing signals.

Third, the error correction speed of 14.1% per month (half-life 4.6 months) indicates that while short-run deviations from long-run equilibrium are gradually corrected, they persist for meaningful durations. This persistence creates window-dressing opportunities for short-term gold traders and complicates the SBV's ability to manage domestic gold prices purely through monetary policy.

6.3 EMH implications from ARIMA results

The ARIMA(0,1,0) selection implies that the Vietnamese gold market satisfies the weak form of the Efficient Market Hypothesis (Fama, 1970): all information in the history of past SJC prices is already incorporated in the current price. This is a stronger statement than the cointegration results, which concern long-run equilibria. The combined findings suggest a market that is informationally efficient in the short run but may exhibit systematic mean-reversion at longer horizons (captured by the ECT).

The Diebold-Mariano test results sharply reject the null of equal forecast accuracy for all three ARIMAX specifications ($DM \approx 4.04$, $p \approx 0.002$). This finding implies that the Vietnamese gold market is not semi-strong form efficient with respect to publicly available exchange rate and world gold price information: incorporating these variables into the forecasting model yields statistically and economically significant improvements over a random walk. The MAPE reduction of 14.9% (from 1.82% to 1.55%) translates into approximately 4.5 million VND per tael in mean absolute forecast error, which is economically meaningful for portfolio managers and traders operating in the Vietnamese gold market.

The combination of weak-form efficiency (random walk univariate model) and semi-strong-form inefficiency (ARIMAX outperforms) is internally consistent: past SJC gold returns contain no information (weak-form), but contemporaneous macro variables — particularly world gold price changes and USD/VND depreciation — are not instantaneously and fully incorporated into SJC prices. This creates a one-period

informational lag that ARIMAX exploits. The finding is consistent with Wang et al. (2021), who document that gold markets in managed-rate economies exhibit partial, not complete, integration with global price signals.

6.4 Structural stability and the SBV reform

Both Chow structural break tests pass comfortably ($p = 0.467$ for COVID; $p = 0.600$ for SBV reform), indicating that the ARDL coefficients are stable across the two regime shifts. This is a substantive result: it implies that the fundamental short-run relationships between SJC gold and macroeconomic variables were not altered by the 2024 auction reform, despite its dramatic effect on the SJC premium level. The reform changed the *level* of the domestic price (captured by D_{sbv24}) without altering the *dynamic* relationships.

7 Conclusion

This paper examines the macroeconomic determinants of domestic SJC gold prices in Vietnam over the period January 2015 to December 2025, combining ARDL bounds testing with ARIMA/ARIMAX out-of-sample forecasting. Our principal findings are as follows.

First, using ARDL bounds testing, we confirm cointegration at the 1% significance level ($F = 5.229$, $p = 0.0002$), supported by a significant error correction term (ECT = -0.141 , $p = 0.016$), with 14.1% of price disequilibria corrected within one month. All four long-run coefficients carry the theoretically predicted signs, although sample-size limitations preclude individual significance.

Second, in the short run, world gold price changes are the dominant driver of SJC price movements, with a contemporaneous elasticity of 0.504 ($p < 0.001$) and a lagged effect of 0.345 ($p < 0.001$). USD/VND exchange rate depreciation exerts a full short-run pass-through ($\beta = 1.039$, $p = 0.027$), consistent with purchasing power parity at the monthly frequency.

Third, ARIMA order selection identifies a random walk as the optimal univariate model, supporting weak-form market efficiency for the Vietnamese gold market. However, ARIMAX models incorporating world gold price changes and the USD/VND exchange rate significantly outperform the random walk baseline ($DM \approx 4.04$, $p < 0.01$), indicating that the market is not semi-strong form efficient with respect to publicly available macro information.

Fourth, structural stability is confirmed across both the COVID-19 shock and the SBV gold market reform, suggesting that the reform altered price *levels* without disrupting the underlying dynamic transmission mechanisms.

Policy implications. For the SBV, the results suggest that exchange rate management has an immediate and near-complete effect on domestic gold prices, implying that episodes of rapid VND depreciation will translate directly into gold price surges. The persistence of the ECT (half-life 4.6 months) indicates a moderate degree of market efficiency that limits the effectiveness of short-term interventions. For investors, the findings confirm gold’s role as an inflation hedge and currency risk hedger in Vietnam, with world gold price movements providing the primary return signal at monthly frequencies.

Limitations and future research. This study faces three limitations. First, the constructed CPI series (annual data compounded to monthly) may understate true monthly inflation volatility. Second, the 2025 SJC price data for some months rely on estimates rather than official records; future work should verify these values. Third, the small sample ($T = 132$) limits the power of long-run inference. Future research could extend the sample horizon, incorporate GARCH-class volatility models to address the documented ARCH effects, or explore nonlinear cointegration methods to examine whether the long-run equilibrium relationship is regime-dependent.

References

- Bahmani-Oskooee, M., & Harvey, H. (2016). Exchange rate sensitivity of U.S. bilateral trade flows. *Economic Modelling*, 53, 329–337.
- Baur, D. G., & McDermott, T. K. (2010). Is gold a safe haven? International evidence. *Journal of Banking & Finance*, 34(8), 1886–1898.
- Derbali, A. (2021). Drivers of gold returns: Evidence from COVID-19 pandemic. *Journal of Public Affairs*, 21(3), e2368.
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Dickey, D. A., & Fuller, W. A. (1981). Likelihood ratio statistics for autoregressive time series with a unit root. *Econometrica*, 49(4), 1057–1072.
- Do, T. T., Tran, T. H., Nguyen, T. T., Le, A. T., & Ninh, L. K. (2023). Is gold an inflation hedge in Vietnam? A non-linear approach. *Cogent Economics & Finance*, 11(2), 2244857. <https://doi.org/10.1080/23322039.2023.2244857>
- Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.
- Hammoudeh, S., & Yuan, Y. (2008). Metal volatility in presence of oil and interest rate shocks. *Energy Economics*, 30(2), 606–620.
- Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291.
- Kremers, J. J., Ericsson, N. R., & Dolado, J. J. (1992). The power of cointegration tests. *Oxford Bulletin of Economics and Statistics*, 54(3), 325–348.
- Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., & Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1–3), 159–178.

- Nguyen, V. H., & Tran, T. Q. (2020). Exchange rate uncertainty and gold prices in Vietnam: A GARCH approach. *Journal of Economics and Development*, 22(1), 45–58.
- Pesaran, M. H., Shin, Y., & Smith, R. J. (2001). Bounds testing approaches to the analysis of level relationships. *Journal of Applied Econometrics*, 16(3), 289–326.
- Phillips, P. C. B., & Perron, P. (1988). Testing for a unit root in time series regression. *Biometrika*, 75(2), 335–346.
- Pierdzioch, C., Risse, M., & Rohloff, S. (2016). Are gold price forecasts more accurate when exchange rates are included? *Empirical Economics*, 50(2), 477–500.
- Rana, F., & O'Connor, F. A. (2023). Domestic macroeconomic determinants of precious metals prices in developed and emerging markets. *International Review of Financial Analysis*, 89, 102813. <https://doi.org/10.1016/j.irfa.2023.102813>
- Sahu, T. N., Mondal, P., & Bandopadhyay, K. (2022). Nexus between gold prices, exchange rates and stock markets. *Journal of Financial Economic Policy*, 14(3), 415–432.
- Salisu, A. A., Raheem, I. D., & Ndako, U. B. (2021). A sectoral analysis of asymmetric nexus between oil price and stock returns. *International Review of Economics & Finance*, 73, 252–265.
- Tran, M. T., Nguyen, T. H., & Le, P. T. (2021). Gold as an inflation hedge in Vietnam: Time-horizon analysis. *Finance Research Letters*, 40, 101683.
- Tully, E., & Lucey, B. M. (2007). A power GARCH examination of the gold market. *Research in International Business and Finance*, 21(2), 316–325.
- Wang, Y., Wei, Y., Wu, C., & Yin, L. (2021). Is gold a safe haven for the dynamic risk of foreign exchange? *Future Business Journal*, 7(1), 56. <https://doi.org/10.1186/s43093-021-00101-9>

World Gold Council. (2023). *Gold Demand Trends Full Year 2022*. London: World Gold Council.

Zivot, E., & Andrews, D. W. K. (1992). Further evidence on the great crash, the oil price shock, and the unit root hypothesis. *Journal of Business & Economic Statistics*, 10(3), 251–270.

A Full ECM Coefficient Table

Table 8 reports all estimated coefficients from the ARDL(1,0,1,1,0) model. Short-run terms not reaching conventional significance thresholds are included for completeness.

Table 8: Full ECM coefficient estimates, ARDL(1,0,1,1,0)

Variable	Coefficient	Std. Error (HAC)	<i>t</i> -stat	<i>p</i> -value
Constant	+1.9834	1.6981	+1.168	0.243
D_{covid}	−0.0088	0.0136	−0.645	0.519
D_{sbv24}	−0.0019	0.0180	−0.107	0.914
D_{tet}	−0.0016	0.0059	−0.265	0.791
ECT ($\ln SJC_{t-1}$)	−0.1410**	0.0586	−2.409	0.016
$\ln EXRATE_{t-1}$	−0.1403	0.1994	−0.704	0.482
$\ln GOLD_{W,t-1}$	+0.1677**	0.0711	+2.358	0.018
$\ln CPI_{t-1}$	+0.1539	0.1201	+1.282	0.200
IR_{t-1}	−0.0044	0.0057	−0.766	0.444
$\Delta \ln SJC_{t-1}$	−0.1340**	0.0634	−2.115	0.035
$\Delta \ln EXRATE_t$	+1.0391**	0.4004	+2.595	0.010
$\Delta \ln GOLD_{W,t}$	+0.5044***	0.0959	+5.260	0.000
$\Delta \ln GOLD_{W,t-1}$	+0.3445***	0.1040	+3.311	0.001
$\Delta \ln CPI_t$	+20.059**	8.860	+2.264	0.024
$\Delta \ln CPI_{t-1}$	−23.361***	7.989	−2.924	0.004
ΔIR_t	+0.0256**	0.0106	+2.409	0.016
<i>N</i>	129	R^2	0.4719	
Adj. R^2	0.4018	AIC	−539.809	

Notes: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$. HAC (Newey-West) robust standard errors throughout. Long-run levels coefficients ($\ln EXRATE_{t-1}$ through IR_{t-1}) combine with ECT to recover the cointegrating vector. Significance of $\ln GOLD_{W,t-1}$ ($p=0.018$) in the levels equation reflects long-run world gold price transmission.

B SBV Refinancing Rate Schedule

Table 9 documents the SBV refinancing rate schedule used in the empirical analysis.

Table 9: SBV refinancing rate schedule, 2015–2025

Effective date	Rate (% p.a.)	Decision number	Direction
January 2015	6.50	Decision 2173/QĐ-NHNN (Oct 2014)	Baseline
July 2017	6.25	Decision 1172/QĐ-NHNN, 30 Jun 2017	Cut
October 2017	6.00	SBV announcement, Sep 2017	Cut
March 2020	5.00	Decision 418/QĐ-NHNN, 16 Mar 2020	Cut (COVID)
May 2020	4.50	Decision 919/QĐ-NHNN, 12 May 2020	Cut
October 2020	4.00	Decision 1728/QĐ-NHNN, 30 Sep 2020	Cut
September 2022	5.00	Decision 1606/QĐ-NHNN, 22 Sep 2022	Hike
October 2022	6.00	Decision 1809/QĐ-NHNN, 24 Oct 2022	Hike
April 2023	5.50	Decision 574/QĐ-NHNN, 31 Mar 2023	Cut
May 2023	5.00	Decision 950/QĐ-NHNN, 23 May 2023	Cut
July 2023	4.50	Decision 1123/QĐ-NHNN, 16 Jun 2023	Cut
December 2025	4.50	Unchanged since Jul 2023	—

Source: State Bank of Vietnam official decisions (QĐ-NHNN), retrieved from sbv.gov.vn.